Qns	Answer	Marks
6(a)(i)	(vertically) downwards	**B1**
6(a)(ii)	magnetic force (on sphere) is always perpendicular to its velocity no work done by force, hence no change in kinetic energy and speed.	**M1** **A1**
6(b)	$mg = Eq$ $E = (1.6 \times 10^{-10} \times 9.81) / (0.27 \times 10^{-9})$ $= 5.8\ \mathrm{N\ C^{-1}}$	**C1** **A1**
6(c)	Magnetic force provides centripetal force $Bqv = mv^2 / r$ $B = (1.6 \times 10^{-10} \times 0.78) / (0.27 \times 10^{-9} \times 3.4) = 0.14\ \mathrm{T}$	**B1** **C1** **A1**